

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	Total Marks:	20 (Answer ANY ONE)
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
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Section / Batch:	Slot-A	Date:	07.08.2026

Code of Conduct

I _____ RASHMI NAURENE _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student: _____

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

1. Derive the number of subcarriers needed to exactly occupy 25 MHz of bandwidth using 15 kHz subcarrier spacing, showing the full calculation. Discuss whether the result divides evenly, and if not, explain how the leftover spectrum is handled as guard band at the channel edges. Describe the purpose this guard band serves in preventing interference with adjacent frequency allocations.



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What We're Trying to Find (Problem Statement)

Q. Derive the number of subcarriers needed to exactly occupy 25 MHz of bandwidth using 15 kHz subcarrier spacing, showing the full calculation. Discuss whether the result divides evenly, and if not, explain how the leftover spectrum is handled as guard band at the channel edges. Describe the purpose this guard band serves in preventing interference with adjacent frequency allocations.

- Imagine you have a radio channel — a chunk of frequency — that's 25 MHz wide. In modern wireless systems (like 4G/5G), this channel isn't used as one big block. Instead, it's chopped up into many small pieces called *subcarriers*, and each subcarrier is 15 kHz wide.
- The question is simple: how many of these 15 kHz pieces can we fit inside our 25 MHz channel?"
- **Given:**
- Total channel width = 25 MHz = 25,000 kHz
- Width of each subcarrier = 15 kHz

2



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The Calculation (Concept)

- "Think of it like slicing a loaf of bread. If the loaf is 25,000 cm long, and each slice is 15 cm wide, how many slices do you get? You just divide the total length by the slice width."
- **Number of subcarriers = Total bandwidth ÷ Subcarrier spacing**
- $25,000 \div 15 = 1666.67$
- "Now here's the catch — you can't cut a bread slice into a fraction and still call it a slice. Same with subcarriers: you can't have two-thirds of one. So we round DOWN, never up (rounding up would mean using more space than we're allowed).
- **Final answer: 1666 subcarriers"**

3



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What Happens to the Leftover Space? (Discussion)

- Since $25,000 \div 15$ didn't come out to a whole number, some space is left over. Let's find out how much.
- Space actually used = $1666 \times 15 \text{ kHz} = 24,990 \text{ kHz} = 24.99 \text{ MHz}$
- Total space we had = 25 MHz
- Leftover = $25 - 24.99 = 10 \text{ kHz}$
- That 10 kHz doesn't just disappear — engineers place it deliberately at the two edges of the channel, like a small empty cushion on either side. This cushion is called a **guard band**."

4



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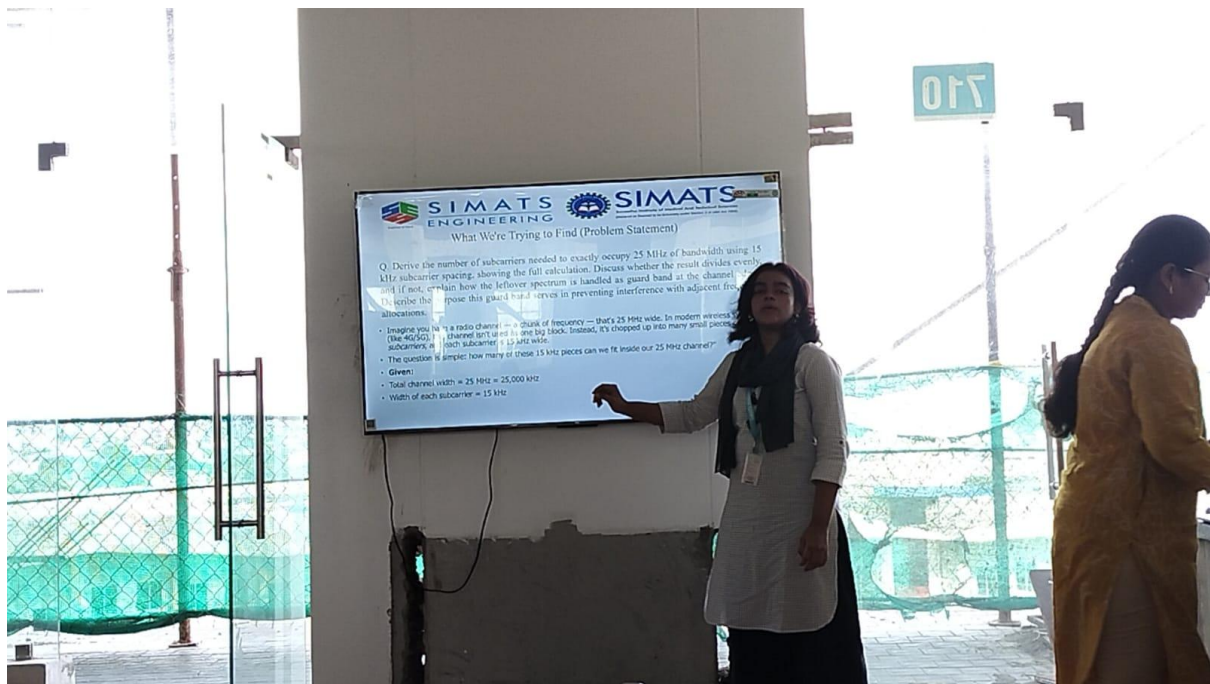
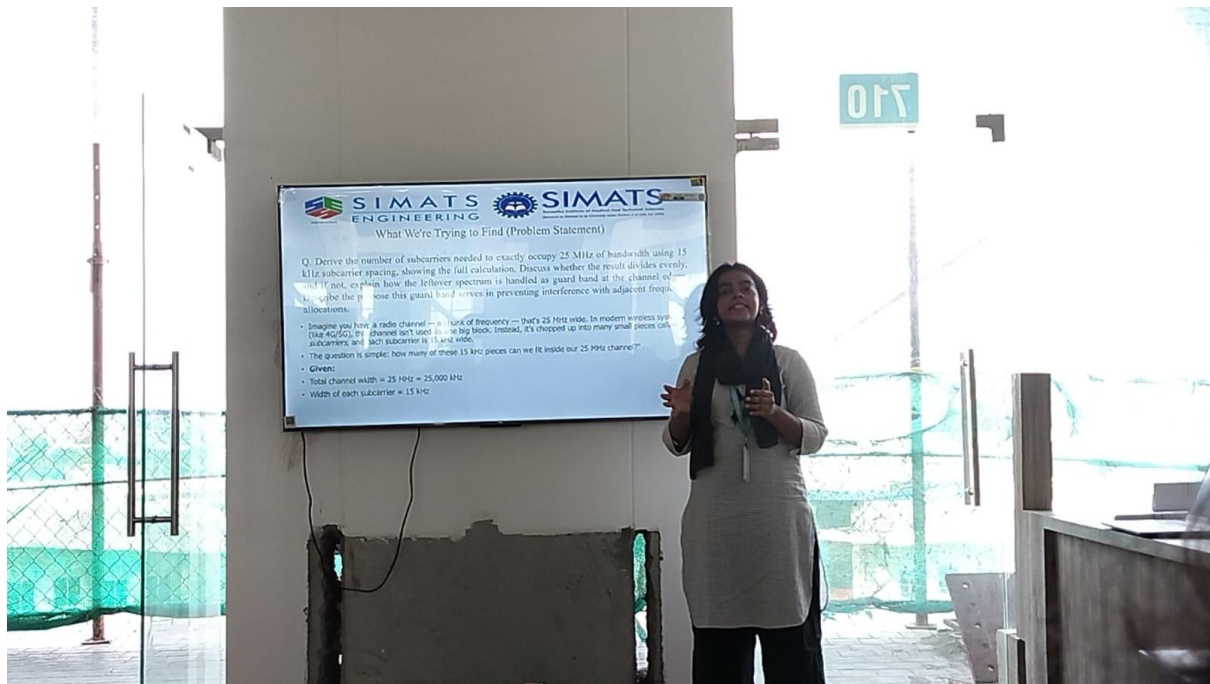


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Why Do We Need This Guard Band? (Conclusion)

- "Think of the guard band like the empty space between parking spots in a parking lot. Without it, cars (or in our case, radio signals) would bump into each other.
- The guard band helps in a few simple ways:
- **Stops signals from leaking into the neighbor's space** — just like your radio signal shouldn't spill into another company's channel next door.
- **Gives room for small timing errors** — devices aren't perfect; their internal clocks drift slightly, and moving phones cause tiny frequency shifts (like how an ambulance siren changes pitch as it passes you). The guard band absorbs these small errors.
- **Handles imperfect equipment** — real filters (the parts that clean up a signal) aren't perfectly sharp-edged; they blur slightly at the boundary, and the guard band gives that blur somewhere safe to exist.
- So the leftover 10 kHz isn't wasted — it's doing quiet, important protective work."

5



**THANK
YOU**